

Global CO₂ Emissions: Trends, Regional Shifts, and Per Capita Inequalities

Scope: 1750–2024 | Data-forward brief for policy analysts and climate negotiators

Sources: Global Carbon Budget 2025 (Our World in Data) and NASA Climate (NOAA Mauna Loa)

Atmospheric CO₂ Reaches 429 ppm — 50% Above Pre-Industrial Levels

429 ppm

February 2026
Mauna Loa Observatory

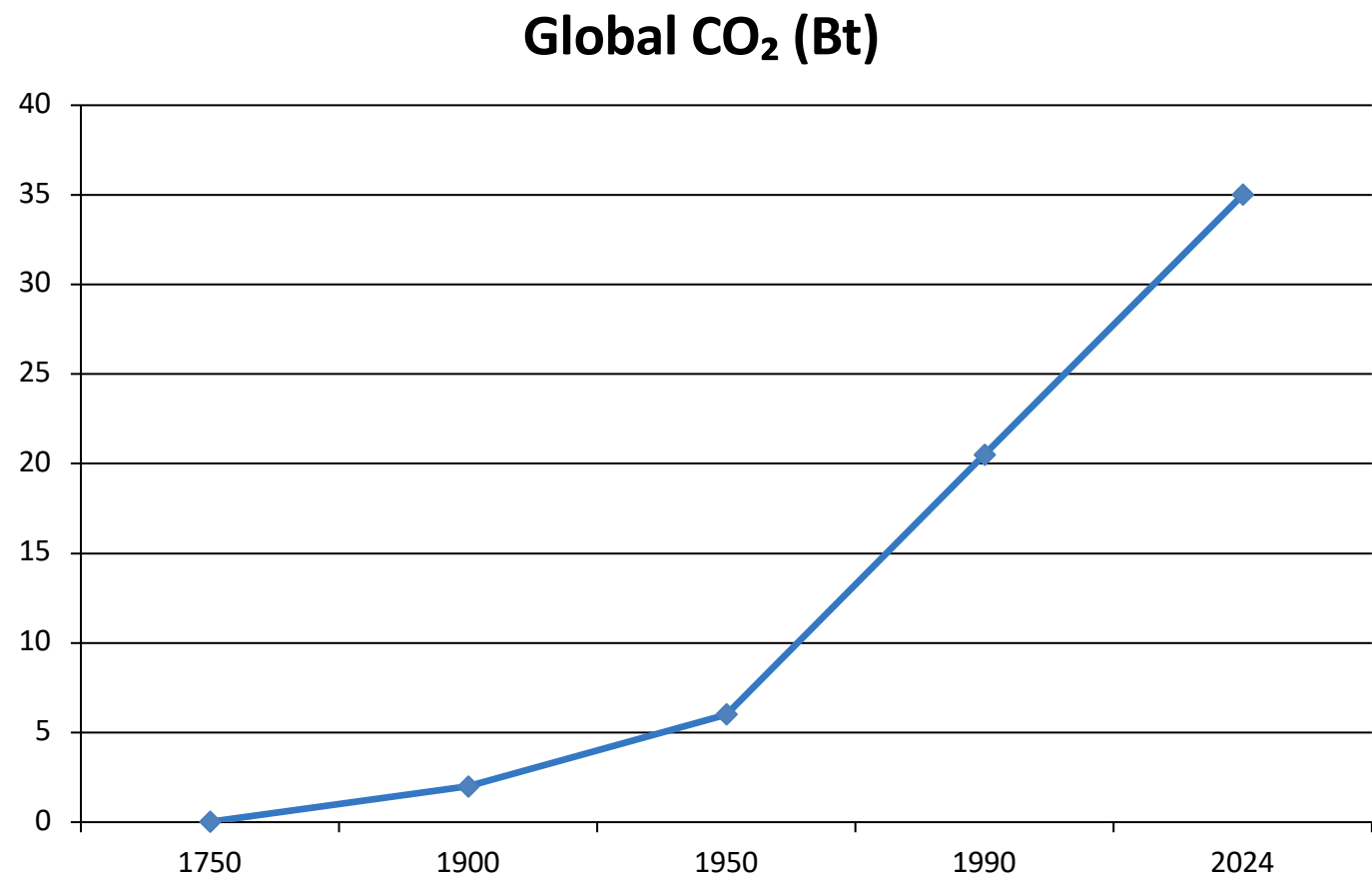
Pre-industrial baseline: ~280 ppm.

Current level is >50% above that baseline.

Comparable natural increase after last ice age took millennia; current rise occurred in ~2 c

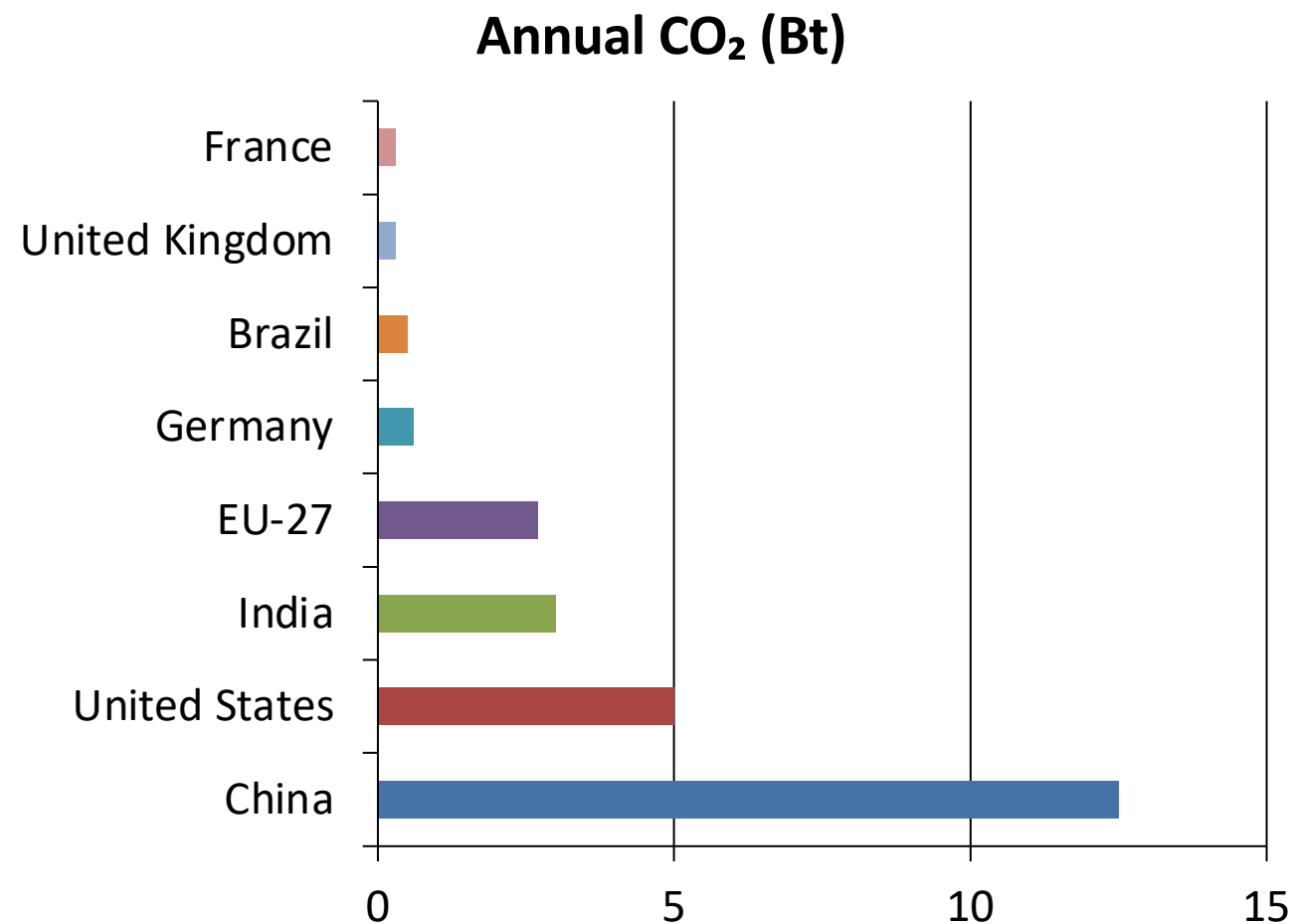
Ground and satellite records show accelerating accumulation tied to fossil fuel combustion

Global Fossil Fuel CO₂ Emissions Grew Six-Fold: 6 Billion to 35+ Billion Ton

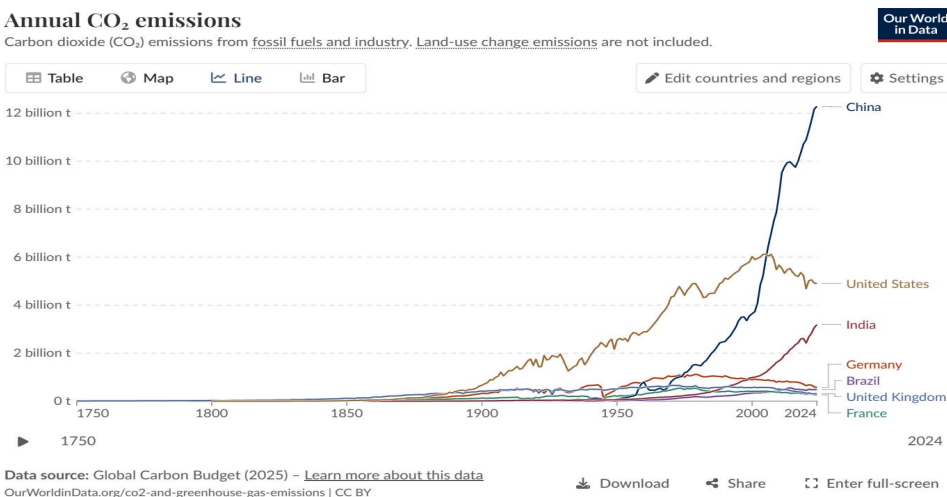


Key inflection points: 1950 (~6 Bt), 1990 (>20 Bt), 2024 (>35 Bt).

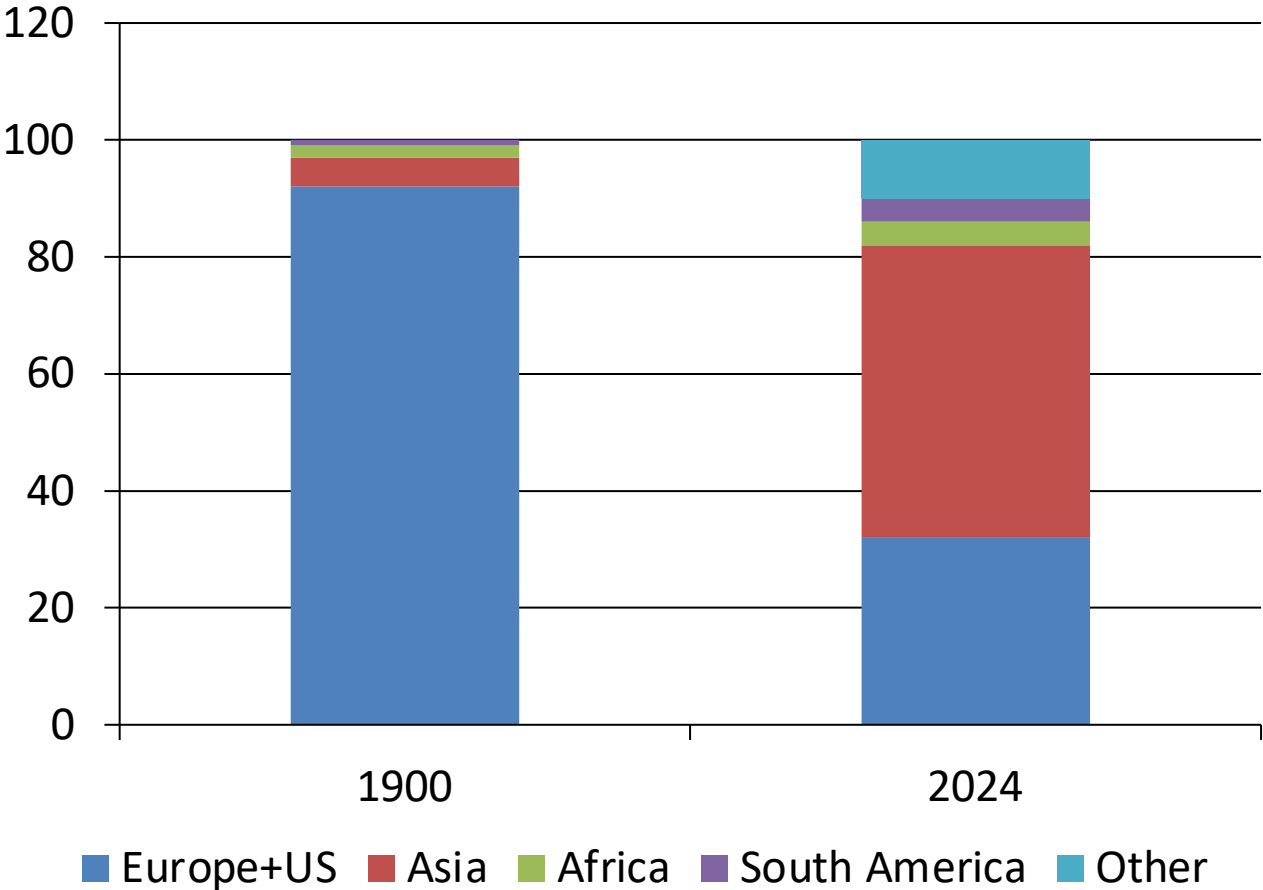
China Now Emits Over 12 Billion Tonnes — More Than Double the United



China: ~12.5 Bt (~27% global), slight increase.
US: ~5 Bt (~14%), down from ~6 Bt peak around 2005.
India: ~3 Bt (~8%), rising rapidly.
EU-27: ~2.7 Bt and declining trend.



Asia Now Accounts for ~50% of Global Emissions; US + Europe Below One

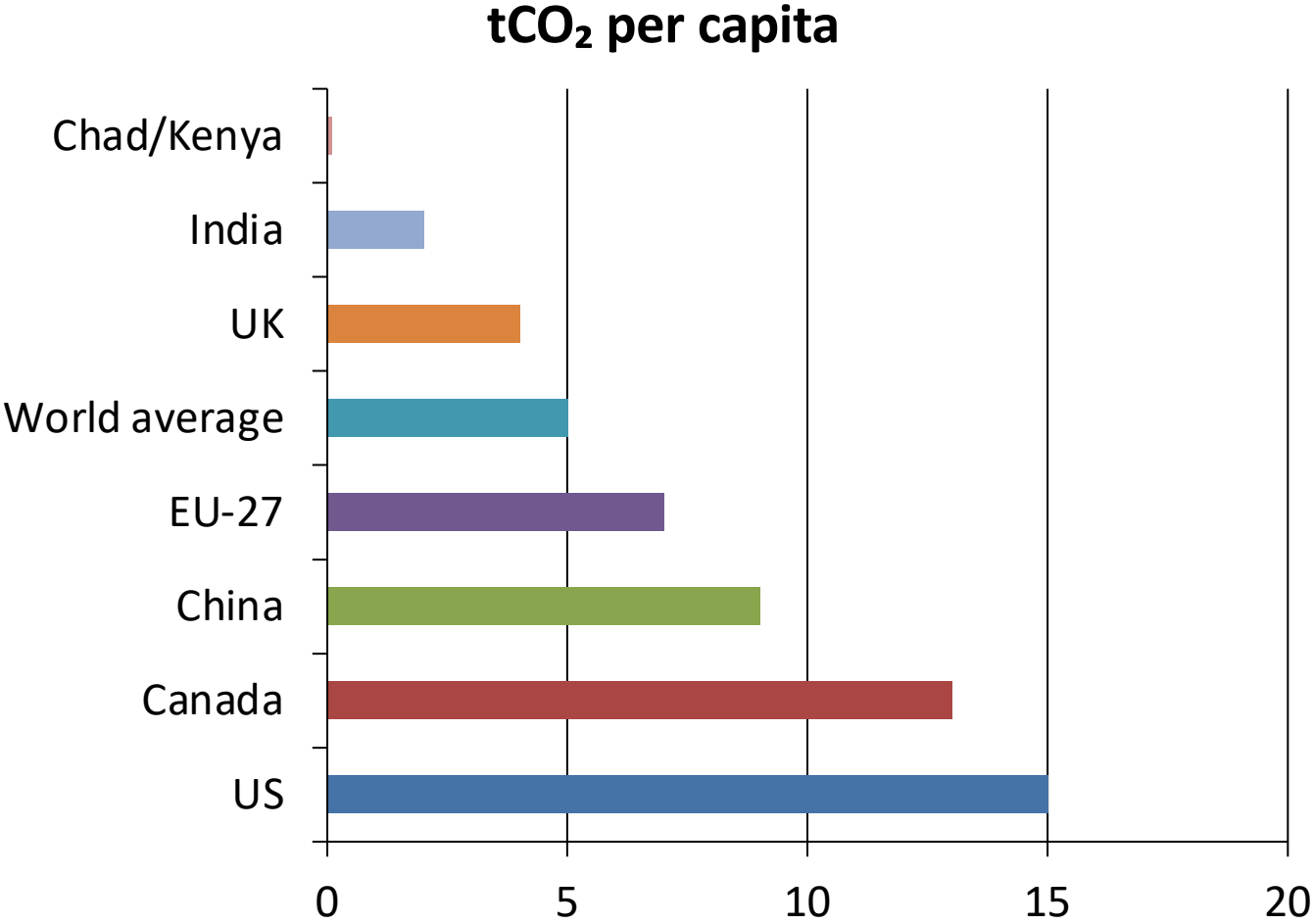


1900: Europe + US >90% of global emissions.

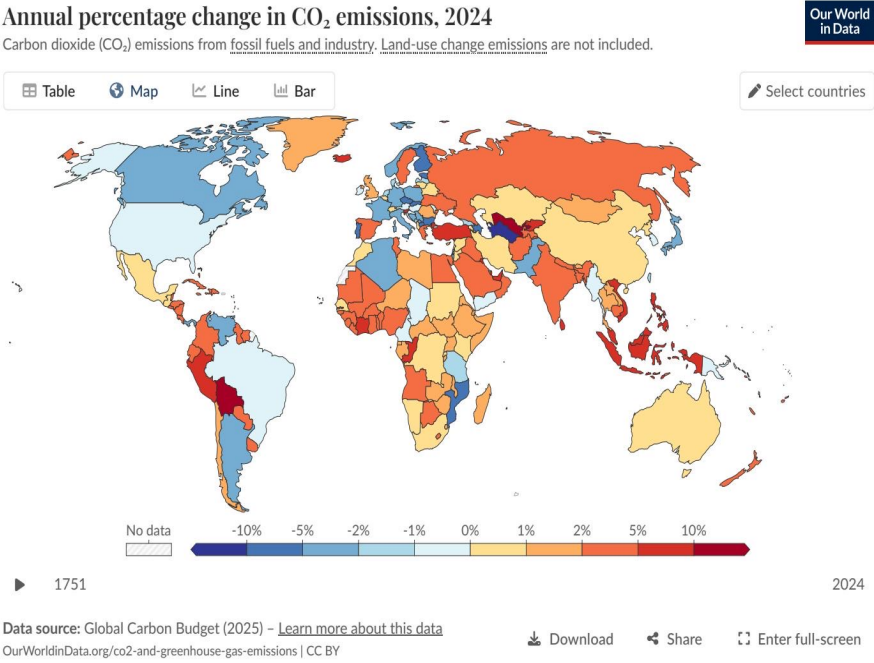
2024: Asia ~50%; Europe + US <33%.

Africa and South America each contribute only 3–4%

Per Capita Gap: 15 t in the US vs. 0.1 t in Chad — a 150× Disparity



Gap highlighted: ~150× between high emitters and low emitters

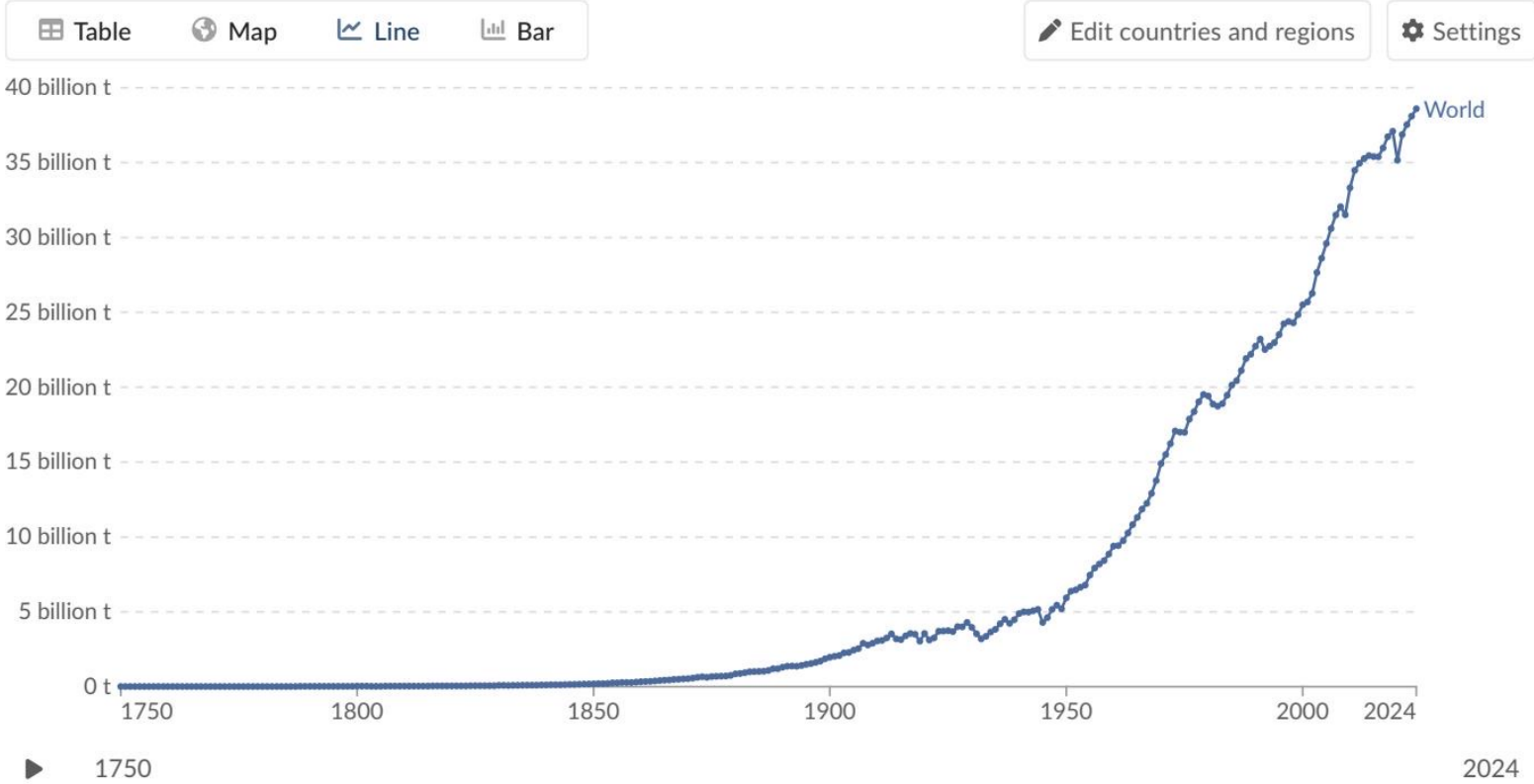


2024 Snapshot: US and Europe Cut Emissions While Russia and SE Asia Inc

Annual CO₂ emissions

Carbon dioxide (CO₂) emissions from fossil fuels and industry. Land-use change emissions are not included.

Our World
in Data



US and much of Europe: negative annual change (≈
Russia, parts of Southeast Asia, and several African
South America shows mixed outcomes, including so

Data source: Global Carbon Budget (2025) – [Learn more about this data](#)
OurWorldinData.org/co2-and-greenhouse-gas-emissions | CC BY

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Energy Mix Explains Why France Emits Far Less Per Capita Than Germany

Similar prosperity does not imply similar emissions.
France and the UK have lower per capita CO₂ than Germany/Netherlands.
Higher nuclear and renewable shares are associated with lower emissions.

Group	Power mix (qualitative)	Per capita CO ₂	Interpretation
France / UK	Higher nuclear + renewables	~4 t	Lower emissions at similar living standards
Germany / Netherlands	Higher fossil fuel share	Higher than France/UK	Power mix drives divergence
World average	Mixed global mix	~5 t	Benchmark for comparison
India	Lower per capita energy use	~2 t	Low today, rising with development

Key Takeaways: Emissions Are Still Rising, But the Map of Responsibility Is

Global fossil CO₂ emissions exceed 35 Bt/year and have not yet peaked.

China contributes over one-quarter of global annual emissions (~12.5 Bt).

Regional center shifted: Europe+US from >90% (1900) to below one-third today; Asia ~50%.

Per capita inequality remains extreme: ~150× gap between ~15 t and ~0.1 t emitters.

Policy and energy mix choices strongly shape per capita outcomes, even at similar income levels.